

Topology First-Year Exam, May 2013, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Prove that the spaces $[0, 1]$ and $[0, 1)$ are not homeomorphic.
2. Let X be a connected normal topological space having more than one point. Prove that X is uncountable.
3. Let $f : X \rightarrow Y$ be a continuous map to a Hausdorff space. Prove that the graph of f , $G = \{(x, f(x)) \mid x \in X\}$, is a closed subset of $X \times Y$.
4. This problem has two parts.
 - (a) Let X be a compact space. Show that for every topological imbedding $f : X \rightarrow Y$ into a Hausdorff space the image $f(X)$ is closed in Y .
 - (b) Suppose that a normal topological space X has the property that for every topological imbedding $f : X \rightarrow Y$, the image $f(X)$ is closed in Y . Does it follow that X is compact?
5. Is every closed subset A of a separable space X separable itself if
 - (a) $X = \mathbb{R}$?
 - (b) $X = \mathbb{R} \times \mathbb{R}$?
 - (c) $X = \mathbb{R}_\ell$?
 - (d) $X = \mathbb{R}_\ell \times \mathbb{R}_\ell$?
6. Does there exist a covering map $p : \mathbb{R}^2 \rightarrow \mathbb{RP}^2$ from the Euclidean plane to the projective plane?

Answer the following with complete definitions or statements or short proofs.

7. State the Tietze Extension Theorem.
8. Is the space $\mathbb{R}^\omega$ connected in the uniform topology?
9. Does there exist a continuous surjective map from the 2-sphere S^2 to the punctured square $([-1, 1] \times [-1, 1]) \setminus \{(0, 0)\}$?
10. Is every connected space path connected?
11. What is a basis of a topology? What is a subbasis?
12. State the Baire Category Theorem.
13. Is the unit circle $S^1 = \{x \in \mathbb{R}^2 \mid \|x\| = 1\}$ a
 - (a) retract of $\mathbb{R}^2$?
 - (b) retract of $\mathbb{R}^2 \setminus \{(2, 0)\}$?
 - (c) deformation retract of $\mathbb{R}^2 \setminus \{(0, 0)\}$?
 - (d) deformation retract of $\mathbb{R}^2 \setminus \{(0, 0), (2, 0)\}$?
 - (e) retract of $\mathbb{R}^2 \setminus \{(0, 0), (2, 0)\}$?

14. State the Brouwer fixed point theorem. Does every continuous map $f : [0, 1] \times [0, 1] \rightarrow [0, 1) \times [0, 1)$ have a fixed point?
15. Can the space of irrationals in the subspace topology be presented as a countable union of nowhere dense subsets?